

Representational Measurement of Similarity: The Additive Difference Model, Revisited

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In dieser Ausarbeitung geht es darum, wie man eine Metrik auf ordinalen Ähnlichkeitsdaten herleiten kann, die zusätzlich additiv ist und durch das additive difference model repräsentiert werden kann. Das Hauptresultat ist, dass die l^p Metrik die einzige additive-difference Metrik, die additiv auf Liniensegmenten ist, was eng mit der Homogenität dieser Metrik zusammenhängt. Zum Herleiten dieses Hauptresultates werden die Grundelemente von Messtheorie, sowie von proximity measurement mithilfe von geometrische Modellen eingeführt. Diese Ausarbeitung basiert auf dem Paper Tversky and Krantz, 1970 und den Büchern Suppes, 2014 und Breinbauer, 1976.

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2 Proximity Structures and Metrics

3 Additive Difference Models and Structures

3.1 Additive Difference Model

3.2 Additive Difference Structure

3.3 Examples

Example 3.1 (p-exp Metric Family).

Let the F in the additive difference model (??) be given by

$$F(t) = q \cdot (e^{t \cdot r} - 1)^p \quad q, r > 0, p \geq 1. \quad (1)$$

The model defined by this F is a metric.

Proof. First, we will prove positivity and symmetry of this function, and then turn to the triangle inequality which is a result of theorem which provides a generalization of the Mikowski inequality.

Part 1 (Positivity and Symmetry).

Positivity and symmetry are both straight forward.

1. Positivity: $\delta(a, b) = 0$ implies $F(|a_i - b_i|) = 0$ for all $i = 1, \dots, n$, which is only possible if $|a_i - b_i| \cdot r = 0$, meaning $a_i = b_i$ for all $i = 1, \dots, n$.
Conversely, if $a \neq b$, then $a_i \neq b_i$ for at least one i in $1, \dots, n$, which implies $F_i(|a_i - b_i|) > 0$, and thus $\delta(a, b) > 0$.
2. Symmetry: Symmetry follows directly from the fact that $|a_i - b_i| = |b_i - a_i|$ holds.

The triangle inequality follows from Mulholland's theorem (Mulholland, 1949, p. 297).

Theorem 3.2 (Mulholland's inequality).

The inequality

$$f^{-1} \left[\sum_{i=1}^n f(|x_i + y_i|) \right] \leq f^{-1} \left[\sum_{i=1}^n f(|x_i|) \right] + f^{-1} \left[\sum_{i=1}^n f(|y_i|) \right].$$

holds for all $x, y \in \mathbb{R}^n, n = 1, 2, \dots$ whenever the following three conditions are satisfied:

1. f is continuous, increasing and $f(0) = 0$.
2. f is convex on $[0, \infty)$.
3. $\log f(e^x)$ is convex for $x \in (-\infty, \infty)$.

Part 1 (Proof that eq. (1) satisfies Mulholland's conditions).

Regarding Point 1: Continuity and $f(0) = 0$ are clear. For monotonicity, note that $e^{rt} - 1$ is increasing, exponentiation is also increasing, and multiplying by $q > 0$ preserves monotonicity. Thus, the composition is also increasing.

Regarding point 2: The same reasoning as for monotonicity applies, except with convexity. The composition of convex functions is convex.

We now proceed to prove the convexity of $\log F(e^t)$. By logarithmic properties, it holds that

$$\log q(e^{re^x} - 1)^p = \log q + p \cdot \log(e^{re^x} - 1).$$

Arbitrary addition and multiplication by $p \geq 1$ preserve convexity, so it suffices to show the convexity of $\log(e^{re^x} - 1)$.

We prove this by examining the second derivative. If the second derivative is strictly positive, the function is convex.

Differentiating twice gives

$$\frac{re^{re^x+x} (r(-e^x) + e^{re^x} - 1)}{(e^{re^x} - 1)^2} \stackrel{!}{>} 0.$$

The denominator is always positive and non-zero, so we can ignore it. Similarly, the factor re^{re^x+x} is always positive and can also be omitted. It remains to show that

$$\begin{aligned} -re^x + e^{re^x} - 1 &> 0 \\ \iff e^{re^x} - re^x &> 1 \end{aligned}$$

holds.

This can be shown in two steps. First, we show that the left-hand side is strictly increasing. Second, we prove that its limit as $x \rightarrow -\infty$ is 1. Therefore, the inequality holds for all $x \in \mathbb{R}$.

1. For the first point, we show that the derivative is greater than zero. Differentiation yields

$$re^x \cdot e^{re^x} - re^x = re^x \cdot (e^{re^x} - 1) > 0$$

since $e^x > 0$.

- 2.

$$\lim_{x \rightarrow -\infty} e^{re^x} - re^x = \lim_{x \rightarrow -\infty} e^{re^x} - \lim_{x \rightarrow -\infty} re^x = 1 - 0 = 1$$

Thus, convexity is established, and the claim follows from Mulholland's theorem.

□

Remark 3.3 (l^p Metrik $\in$ p-exp Metric Family). If we set $q = r^{-p}$ in (1), we obtain the Minkowski metric in the limit as $r \rightarrow 0$.

Proof. Setting $q = r^{-p}$ yields

$$r^{-p}(e^{rt} - 1)^p = \left(\frac{e^{rt} - 1}{r} \right)^p.$$

As $r \rightarrow 0$, both the numerator and denominator tend to 0 (we can pull the limit inside the exponent since the exponentiation is continuous), therefore we can apply L'Hôpital's rule. Hence

$$\begin{aligned} \lim_{r \rightarrow 0} \left(\frac{e^{rt} - 1}{r} \right)^p &= \left(\lim_{r \rightarrow 0} \frac{e^{rt} - 1}{r} \right)^p && \text{(continuity)} \\ &= \left(\lim_{r \rightarrow 0} \frac{t \cdot e^{rt} - 1}{1} \right)^p && \text{(L'Hôpital's rule)} \\ &= t^p. \end{aligned}$$

□

Das ist von Bedeutung, da dies mithilfe des Hauptresultates ?? zeigt, dass die Metriken der p -exp Familie keine Metriken mit additiven Segmenten sein können, obwohl man die l^p Metrik durch Grenzwertbildung aus dieser Familie darstellen kann. Das bedeutet, dass im Grenzwertprozess aus der Eigenschaft "additiv auf den Koordinatenachsen", die Eigenschaft "additiv auf beliebigen Liniensegmenten" wird.

Not every function that can be represented by the additive difference model is a metric.

4 Segmental Additivity

4.1 Segmentally Additive Metrics

4.2 Segmentally Additive Proximity Structures

4.3 Examples

5 Combining Segmental Additivity with the AD-Model

5.1.1 Convexity of δ Balls

6 Discussion

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